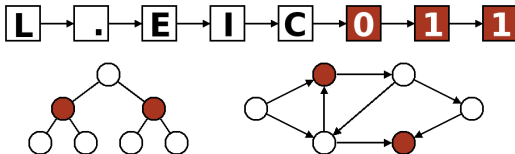


Introduction

L.EIC

Algorithms and Data Structures

2026/2027



AP Tomás P Ribeiro

General Information

- Main Resources:

Algorithms and Data Structures L.E.I.C
2026/2027 Edition
Bachelor in Informatics and Computing Engineering

Lectures
Ana Paula Tomás
Pedro Ribeiro

- Lectures (theoretical classes) start on 17 Sep
- Practical classes start on the 21-25 Sep week



The L.E.I.C logo consists of the letters L, E, I, C in a sequence, with the 'E' and 'I' having small circles below them. To the right of the logo are two graph diagrams. The first is a simple tree structure with a root node and three leaf nodes. The second is a more complex graph with six nodes and several edges, including a cycle.

► **Webpage:** course webpage (classes, slides and all relevant information)
<https://www.dcc.fc.up.pt/~pribeiro/aulas/aed2627/>



Discord: communication/discussion ("*instant messaging*" format)
(*invite already sent by email*)

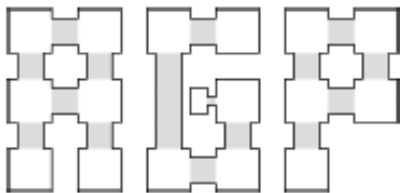


Mooshak: Code submissions with automatic test evaluation
<https://mooshak.dcc.fc.up.pt/~aed/>
(*passwords will be sent by email*)

Teaching Staff

- **Theoretical Classes** (*one 2h class per week*)
 - ▶ Ana Paula Tomás (FCUP) - *1st Half of Semester*
 - ▶ Pedro Ribeiro (FCUP) - *2nd Half of Semester*
- **Practical Classes:** (*one 2h class per week*)
 - ▶ Ana Paula Tomás (FCUP)
 - ▶ Filipa Ramos (FEUP)
 - ▶ Flávia Carvalhido (FEUP)
 - ▶ Jadna Cruz (FEUP)
 - ▶ Jorge Oliveira (FEUP)
 - ▶ Lázaro Costa (FEUP)
 - ▶ Patrick Daniel (FCUP)
 - ▶ Pedro Ribeiro (FCUP)
 - ▶ Tiago Carvalho (FEUP)
 - ▶ Vanessa Silva (FCUP)

Teaching Staff - Ana Paula Tomás



Name: Ana Paula Tomás

Office: FC6 1.15 (at DCC/FCUP - Campo Alegre)

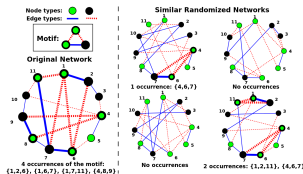
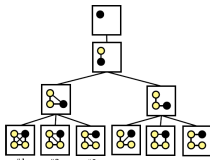
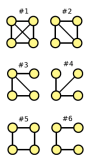
Website: <https://www.dcc.fc.up.pt/~apt/>

Course Director of L:CC

Main Research Interests

- Design and Analysis of Algorithms
- Geometric and Network Problems
- Discrete Optimization and Constraint Programming
- Enumerative Combinatorics
- Computer Assisted Learning

Teaching Staff - Pedro Ribeiro



Name: Pedro Ribeiro

Office: FC6 1.47 (DCC/FCUP)

Website: <https://www.dcc.fc.up.pt/~pribeiro/>

Course Director of M:CC

Main Research Interests

- Complex Networks, Network Science, Graphs, Data Science/ML/AI
- Algorithms and Data Structures, Complexity
- Parallel and Distributed Computing
- Computer Science Education and Programming Contests

Pedro Ribeiro - Competitive Programming

I am involved in many **algorithmic programming contests**
(organization, creating and solving problems, student mentoring and training, ...)

- Pre-University Students (Primary and Secondary Education)
 - ▶ **Olympiads** in Informatics (National and International, e.g., ONI, IOI)
 - ▶ **Bebras** - International Computational Thinking Challenge
- University Students
 - ▶ National and International **ICPC** Contests (e.g., MIUP, SWERC, EUC)



Pre-requirements

- Students should have basic knowledge of programming and **C++**

```
// Your first C++ Program

#include <iostream>

int main() {
    std::cout << "Hello World!" << std::endl;
    return 0;
}
```

Hopefully not really your FIRST program 😊

Theoretical classes

- on site (*"face to face"* classes)
- (formal) exposition of the subjects, presentation of examples, analysis and discussion, usage of whiteboard, some livecoding

Practical classes

- pen and paper exercises (*you will have a written exam*)
- programming exercises in **C++** (*we will use C++17*)
- no imposed IDE (suggestion: VSCode, CLion)
- automatic evaluation (Mooshak and test cases)

Evaluation

Final mark is computed up from:

- **NP:** practical grade, obtained with 2 onsite individual mooshak practical tests (10 points each) **[not repeatable on "recurso"!]**
- **E:** exam grade ("época normal" or "recurso"), obtained with a written exam with a grade from 0 to 20.

Two possible weightings

Your grade will be the **maximum** between:

- 65% for written exam + 35% practical grade
or
- 75% for written exam + 25% practical grade



$$\text{Grade} = \max(E \times 0.65 + NP \times 0.35, E \times 0.75 + NP \times 0.25) \geq 9.5$$

Elegibility for the Exam

Attending practical classes

Students may not exceed the limit of absences (**25%** of TP classes)

Using last year "frequency"

Students who have not exceeded the absence limit on the previous academic year are **exempt** from TP classes, **but they are advised to attend them.**

Keeping last year practical tests grade

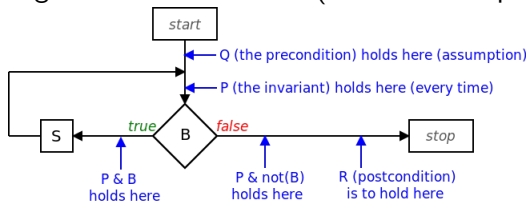
Students **who did not pass the course in 2025/2026** and completed the practical component that academic year, **may keep the grade of their practical tests, if they wish.** They must inform the lecturers at the beginning of the academic year.

Objectives

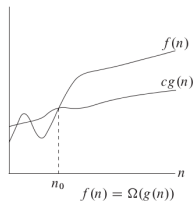
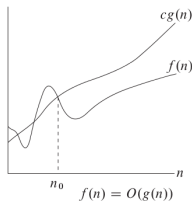
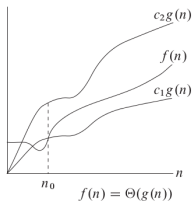
- Analyze the **correctness of simple algorithms**
- Analyze the **temporal and spatial complexity of algorithms**
- Understand the concept of **abstract data type** and know how to organize programs around this concept
- Know the **fundamental data structures and associated algorithms** and respective complexity
- **Choose appropriate** collections, data structures and algorithms to solve practical problems
- **write programs in C++** that implement and use the fundamental data structures and algorithms

An overview of the program

- Algorithmic **correctness** (and the concept of **loop invariant**)

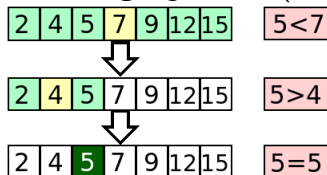


- Algorithmic **efficiency** (time and space)
(asymptotically analysis and the **Big O** notation)

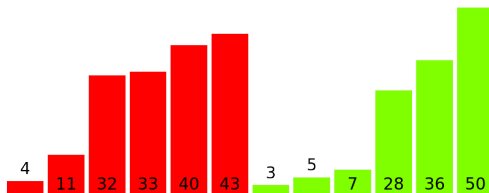


An overview of the program

- **Searching** algorithms (linear search, binary search and variants)

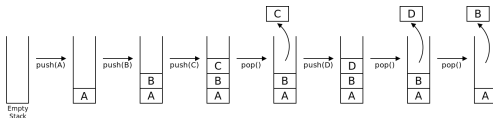
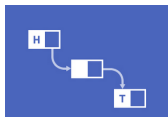


- **Sorting** algorithms (comparison based sorting and linear sorting)
[e.g. MergeSort, QuickSort, RadixSort, CountingSort, ...]

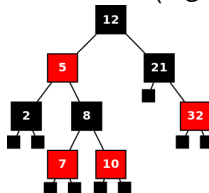
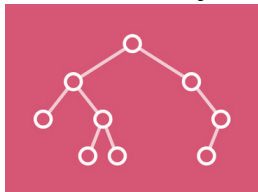


An overview of the program

- Linear data structures: **lists**, **stacks**, **queues**

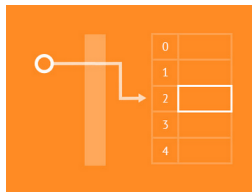


- Hierarchical data structures: **binary trees**, **binary search trees**, **balanced binary trees** and variants (*e.g. AVL and Red-Black trees*)



An overview of the program

- Other essential data structures (**priority queues, hash tables**)



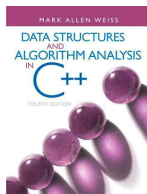
- Graphs** (concept, representation, traversal and fundamental algorithms)
(e.g. *DFS, BFS and applications*)



Bibliography - Main Books

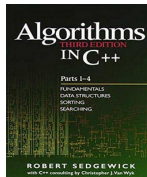
- **Data Structures & Algorithm Analysis in C++**

Mark Allen Weiss, 4th Edition, Pearson Education



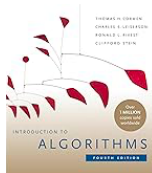
- **Algorithms in C++**

Robert Sedgewick, 3rd Edition, Princeton University



- **Introduction to Algorithms**

TH Cormen, CE Leiserson, RL Rivest and C Stein
4th Edition, MIT Press (also known as **CLRS**)



Don't outsource the thinking!

- **Think first:** what is the problem? what is my approach to it?
- **Ask for help:** Ask for explanations and hints, not for solutions
(and you can ask to a professor, even on discord)
- **Critique:** Is the answer correct? Why? What is the complexity?
(don't trust on "sounding" convincing, challenge and look for counterexamples)

If you use AI, use it to learn, not to think for you!

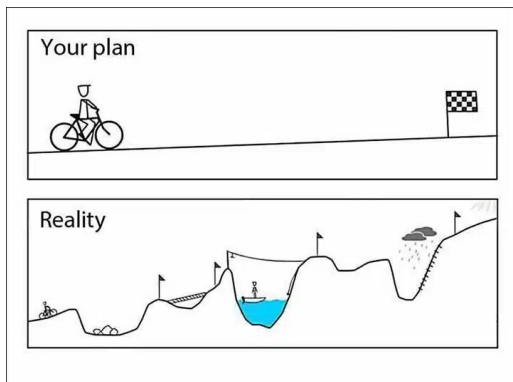
In this course, **your goal isn't to produce code.**
It's to learn how to **think algorithmically** and to **get better at DSA.**

- If you **focus on this goal (learning)**, you will understand why simply copying solutions is not the answer and will not help you
- If AI does the thinking, you only get an "answer". If you do the thinking and AI helps you, **you learn and improve your DSA knowledge!**

(DSA: Data Structures and Algorithms)

Enjoy the semester and this course!

Genius is 1% inspiration and 99% perspiration (Thomas Edison)



- The exercises are designed to help you grow: the solution is only the "artifact", the journey you go through to get them is vital
- Don't be afraid to make errors: learn with them! (*Super Mario Effect*)